



$$H_i^{i-1} = \begin{bmatrix} \cos(\theta_i) & -\sin(\theta_i) \cos(\alpha_i) & \sin(\theta_i) \sin(\alpha_i) & a_i \cos(\theta_i) \\ \sin(\theta_i) & \cos(\theta_i) \cos(\alpha_i) & -\cos(\theta_i) \sin(\alpha_i) & a_i \sin(\theta_i) \\ 0 & \sin(\alpha_i) & \cos(\alpha_i) & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$H_1^0 = \begin{bmatrix} \cos(q_0) & 0 & -\sin(q_0) & 0 \\ \sin(q_0) & 0 & \cos(q_0) & 0 \\ 0 & -1 & 0 & H_0 \\ 0 & 0 & 0 & 1 \end{bmatrix}; \quad H_2^1 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & L_1 + q_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$H_3^2 = \begin{bmatrix} 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & L_2 + q_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}; \quad H_4^3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & L_3 + q_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$H_4^0 = H_1^0 H_2^1 H_3^2 H_4^3$$

$$H_2^0 = H_1^0 H_2^1 = \begin{bmatrix} \cos(q_0) & 0 & -\sin(q_0) & 0 \\ \sin(q_0) & 0 & \cos(q_0) & 0 \\ 0 & -1 & 0 & H_0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & L_1 + q_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & \sin(q_0) & \cos(q_0) & -\sin(q_0)(L_1 + q_1) \\ 0 & -\cos(q_0) & \sin(q_0) & \cos(q_0)(L_1 + q_1) \\ 1 & 0 & 0 & H_0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$H_4^2 = H_3^2 H_4^3 = \begin{bmatrix} 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & L_2 + q_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & L_3 + q_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & -1 & -L_3 - q_3 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & L_2 + q_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$H_4^0 = \begin{bmatrix} \sin(q_0) & -\cos(q_0) & 0 & \cos(q_0)(L_2 + q_2) - \sin(q_0)(L_1 + q_1) \\ -\cos(q_0) & -\sin(q_0) & 0 & \cos(q_0)(L_1 + q_1) + \sin(q_0)(L_1 + q_1) \\ 0 & 0 & -1 & H_0 - L_3 - q_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Pour $[q_0, q_1, q_2, q_3] = [\pi/2, 0.5, 0.5, 0.5]$; $L_1 = 1$; $L_2 = 1$; $L_3 = 1$; $H_0 = 2$;

$$H_4^0 = \begin{bmatrix} 1 & 0 & 0 & -1.5 \\ 0 & -1 & 0 & 1.5 \\ 0 & 0 & -1 & 1.5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Exercice 2

1- Robot 6R

Représentation de repères selon **DH standard**

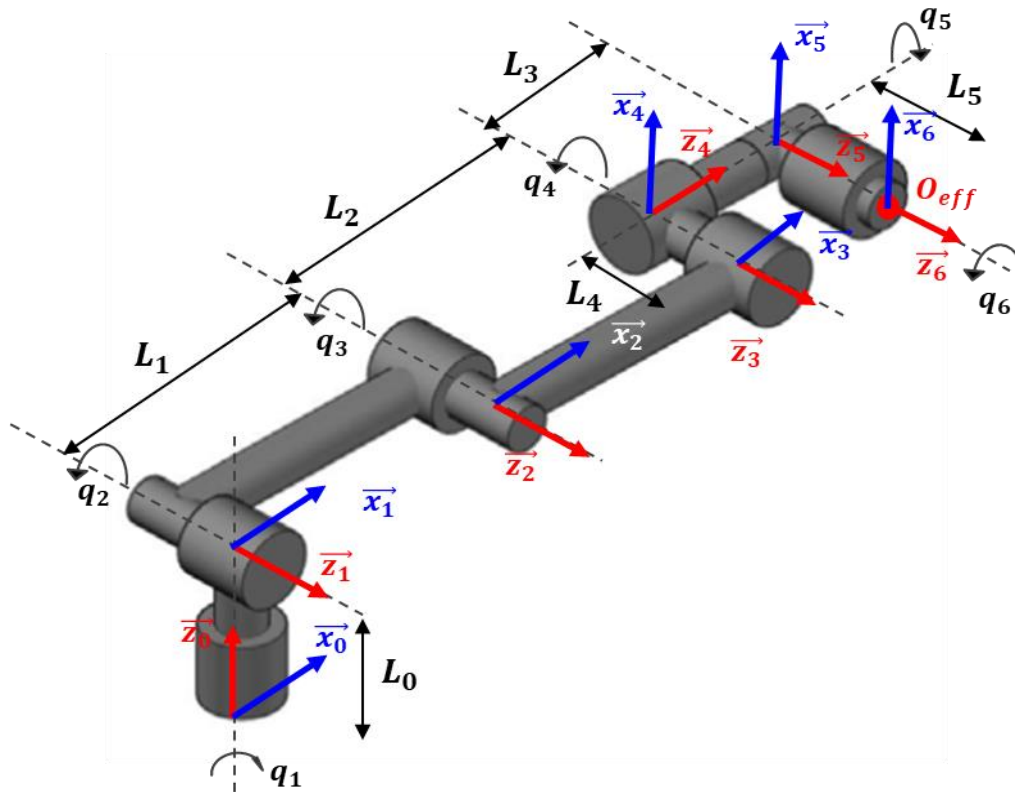


Table des paramètres de DH standard

i	θ_i	d_i	a_i	α_i
1	q_1	L_0	0	$-\frac{\pi}{2}$
2	q_2	0	L_1	0
3	q_3	0	L_2	0
4	$\frac{\pi}{2} + q_4$	L_4	0	$\frac{\pi}{2}$
5	q_5	L_3	0	$-\frac{\pi}{2}$
6	q_6	L_5	0	0

2- Robot manipulateur ACMA-H80

Représentation de repères selon **DH standard**

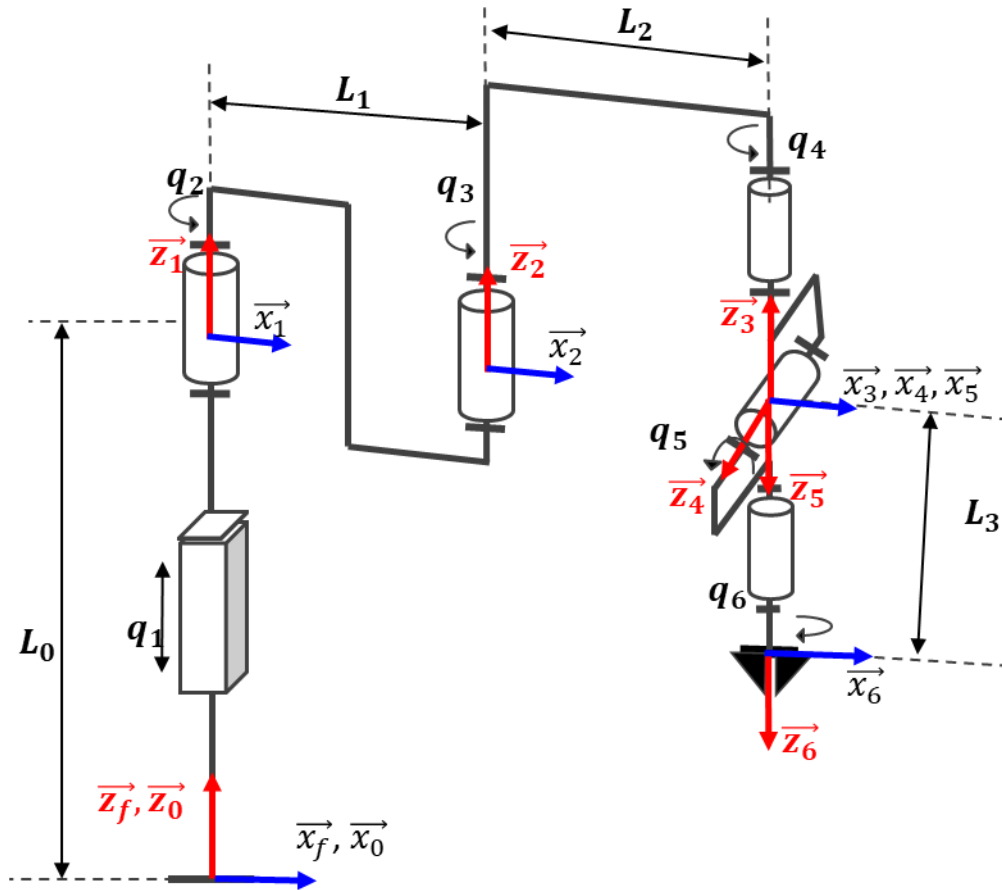


Table des paramètres de DH standard

i	θ_i	d_i	a_i	α_i
1	0	$L_0 + q_1$	0	0
2	q_2	0	L_1	0
3	q_3	0	L_2	0
4	q_4	0	0	$\frac{\pi}{2}$
5	q_5	0	0	$\frac{\pi}{2}$
6	q_6	L_3	0	0

